Mahad Sharfi

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# Question 1: Day Number Program

C# program that accepts a day number from 1 to 7 and prints the corresponding day name. If the input is outside this range, it prints “Invalid number.”

## C# Source Code

```
■using System;

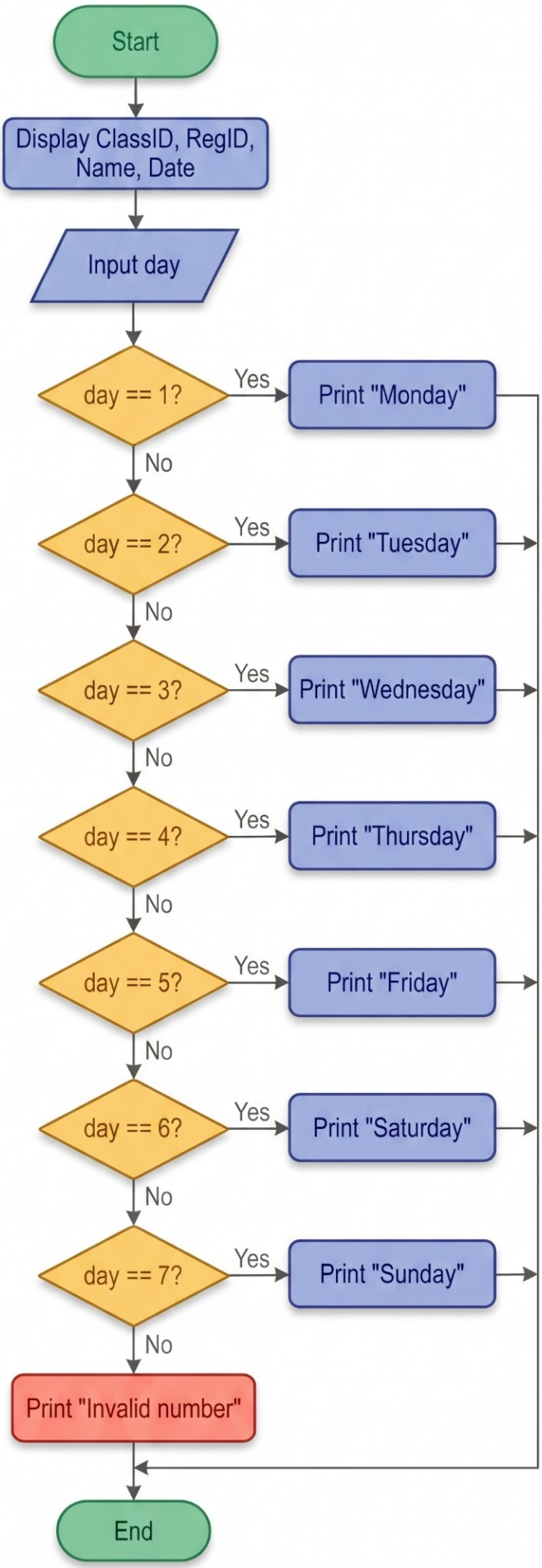
namespace Question1
{
    class Question1
    {
        static void Main(string[] args)
        {
            int day;


            //We have to enter the day number (1 to 7)
            Console.WriteLine("Enter Day Number (1-7)");
            day = Convert.ToInt32(Console.ReadLine());

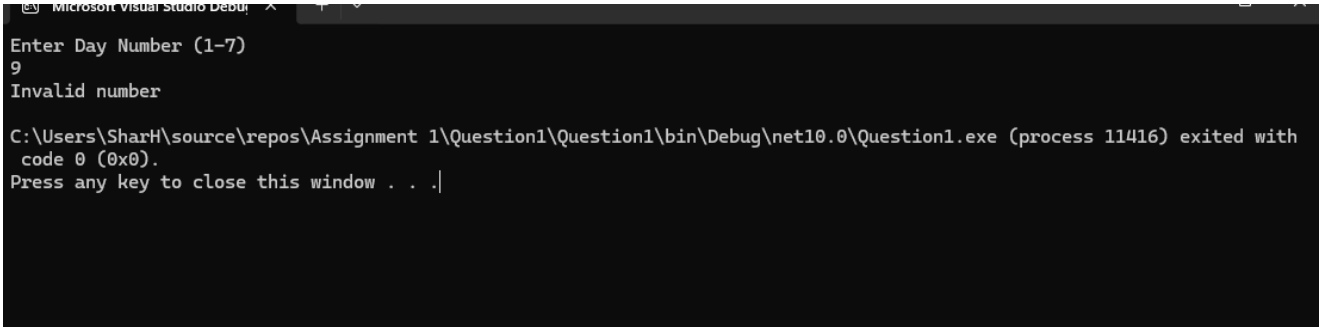

            //We have to check the day number and print day name
            if (day == 1)
            {
                Console.WriteLine("Monday");
            }
            else if (day == 2)
            {
                Console.WriteLine("Tuesday");
            }
            else if (day == 3)
            {
                Console.WriteLine("Wednesday");
            }
            else if (day == 4)
            {
                Console.WriteLine("Thursday");
            }
            else if (day == 5)
            {
                Console.WriteLine("Friday");
            }
            else if (day == 6)
            {
                Console.WriteLine("Saturday");
            }
            else if (day == 7)
            {
                Console.WriteLine("Sunday");
            }

            //If the number is not between 1 and 7
            else
            {
                Console.WriteLine("Invalid number");
            }
        }
    }
}
```

# Flowchart


Start
Display ClassID, RegID, Name, Date
Input day
day == 1?
Yes
Print "Monday"
No
day == 2?
Yes
Print "Tuesday"
No
day == 3?
Yes
Print "Wednesday"
No
day == 4?
Yes
Print "Thursday"
No
day == 5?
Yes
Print "Friday"
No
day == 6?
Yes
Print "Saturday"
No
day == 7?
Yes
Print "Sunday"
No
Print "Invalid number"
End

# Program Output: Invalid Input



# Program Output: Valid Input



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# Question 2: Shop Discount

The program calculates a discount based on the purchase amount, then displays the discount percentage, discount amount, and final payable amount.

## C# Source Code

```
■using System;

namespace ConsoleApplication2
{
    class Question2
    {
        static void Main(string[] args)
        {
            int amount;
            int discountPercentage;
            int discountAmount;
            int finalAmount;


            //We have to enter the purchase amount (Digits)
            Console.WriteLine("Enter Purchase Amount");
            amount = Convert.ToInt32(Console.ReadLine());

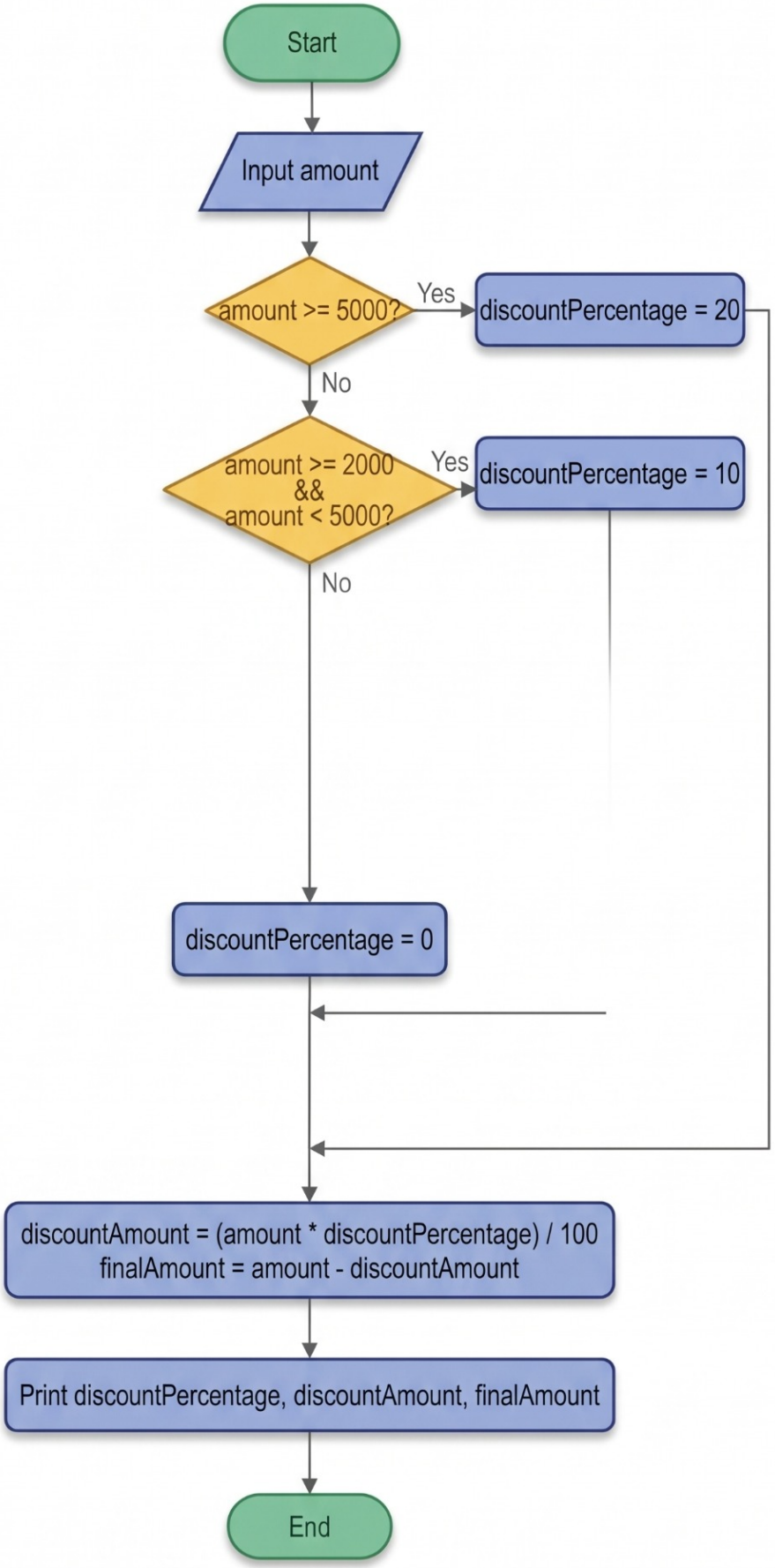

            //We have to check the amount for discount percentage
            if (amount >= 5000)
            {
                discountPercentage = 20;
            }
            else if (amount >= 2000 && amount < 5000)
            {
                discountPercentage = 10;
            }
            else
            {
                discountPercentage = 0;
            }


            //We have to calculate the discount amount and final payable amount
            discountAmount = (amount * discountPercentage) / 100;
            finalAmount = amount - discountAmount;

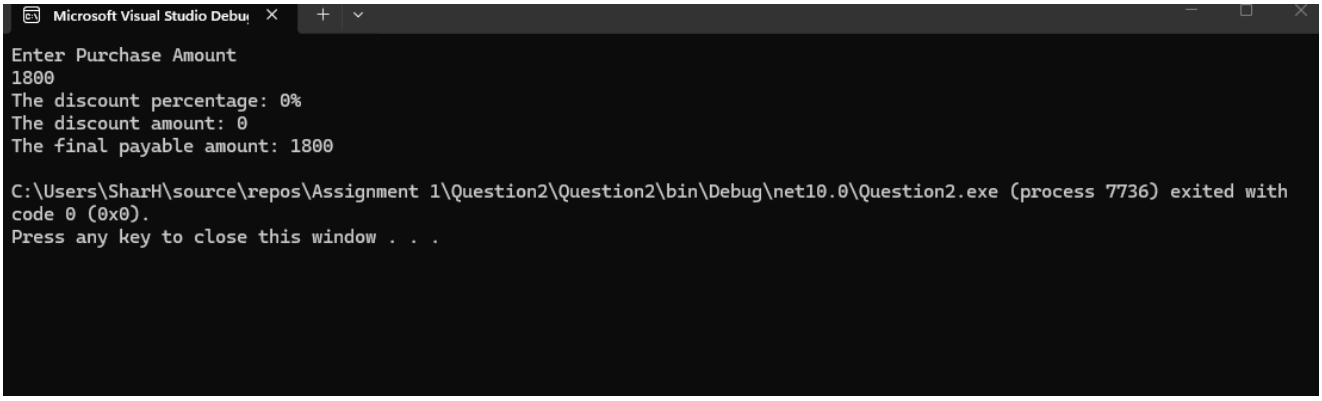

            //We have to print the final results
            Console.WriteLine("The discount percentage: " + discountPercentage + "%");
            Console.WriteLine("The discount amount: " + discountAmount);
            Console.WriteLine("The final payable amount: " + finalAmount);
        }
    }
}
```

# Flowchart

# Question 2 (Shop Discount)



# Program Output: 20% Discount



# Program Output: 0% Discount



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# Question 3: Leap Year

The C# program accepts a year and prints whether it is a leap year or not.

## C# Source Code

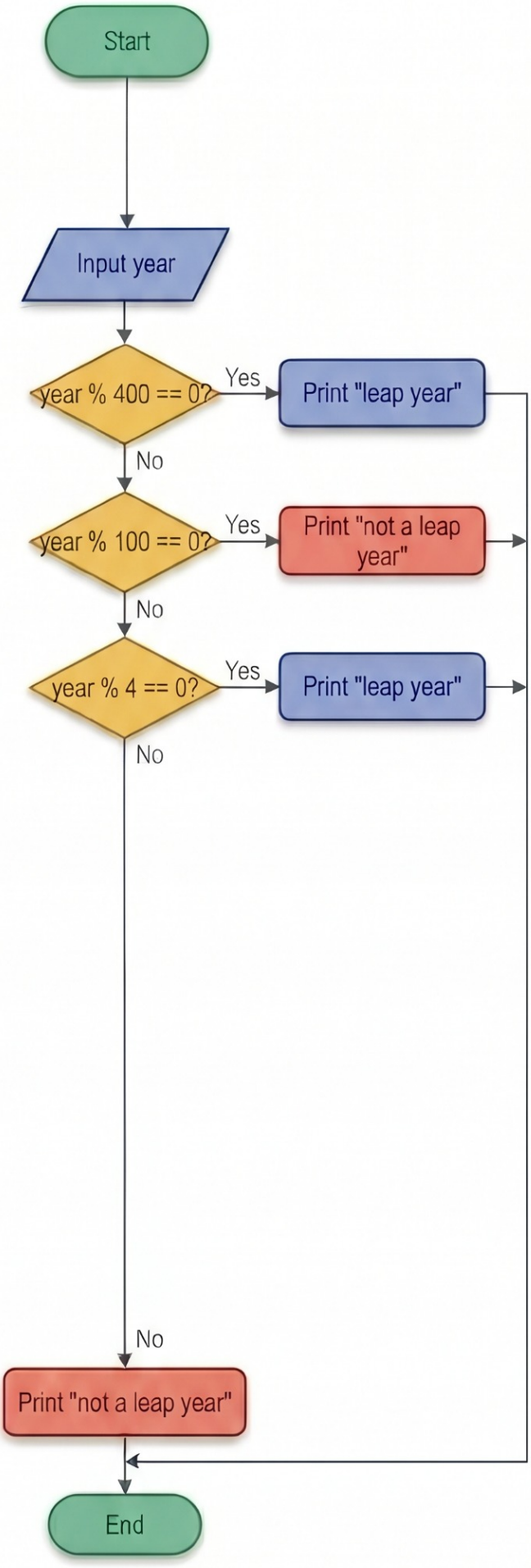
```
using System;

namespace ConsoleApplication3
{
    class Question3
    {
        static void Main(string[] args)
        {
            int year;

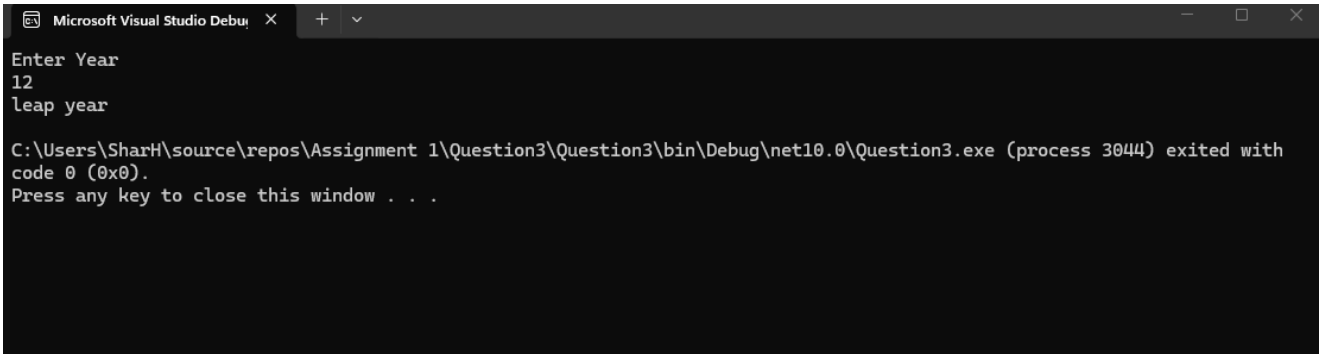
            //We have to enter the year (Digits)
            Console.WriteLine("Enter Year");
            year = Convert.ToInt32(Console.ReadLine());


            //We have to check if the year is a leap year or not
            if (year % 400 == 0)
            {
                Console.WriteLine("leap year");
            }
            else if (year % 100 == 0)
            {
                Console.WriteLine("not a leap year");
            }
            else if (year % 4 == 0)
            {
                Console.WriteLine("leap year");
            }
            else
            {
                Console.WriteLine("not a leap year");
            }
        }
    }
}
```

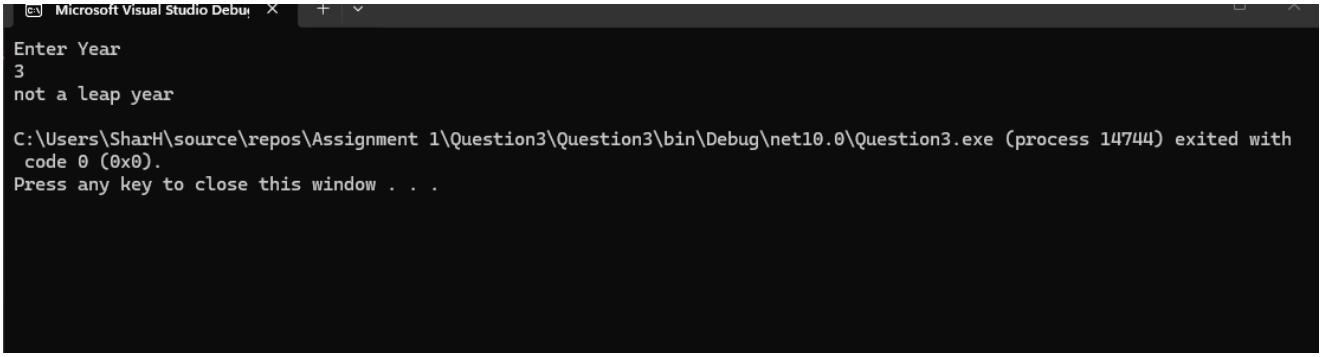
# Flowchart


Start
Input year
year % 400 == 0?
Yes
Print "leap year"
No
year % 100 == 0?
Yes
Print "not a leap year"
No
year % 4 == 0?
Yes
Print "leap year"
No
No
Print "not a leap year"
End

# Program Output: Leap Year



# Program Output: Not a Leap Year



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# Question 4: Calculator

The C# program takes two numbers and an operator (+, -, *, /), performs the selected arithmetic operation, and displays the result. For an unsupported operator, it displays an error message.

## C# Source Code

```
■using System;

namespace ConsoleApplication4
{
    class Question4
    {
        static void Main(string[] args)
        {
            int a, b, sum, sub, mul, div;
            string sign;

            //We have to type the first number
            Console.WriteLine("Enter First Number");
            a = Convert.ToInt32(Console.ReadLine());

            //We have to type the Second number
            Console.WriteLine("Enter Second Number");
            b = Convert.ToInt32(Console.ReadLine());

            //now we use arthmetic operations to perform a calculations
            Console.WriteLine("Enter Sign (+, -, *, /)");
            sign = Console.ReadLine();

            //this (if) is used for the addition operation whenever we wants to add the numbers
            if (sign == "+")
            {
                sum = a + b;
                Console.WriteLine("Addition is:" + sum);
            }

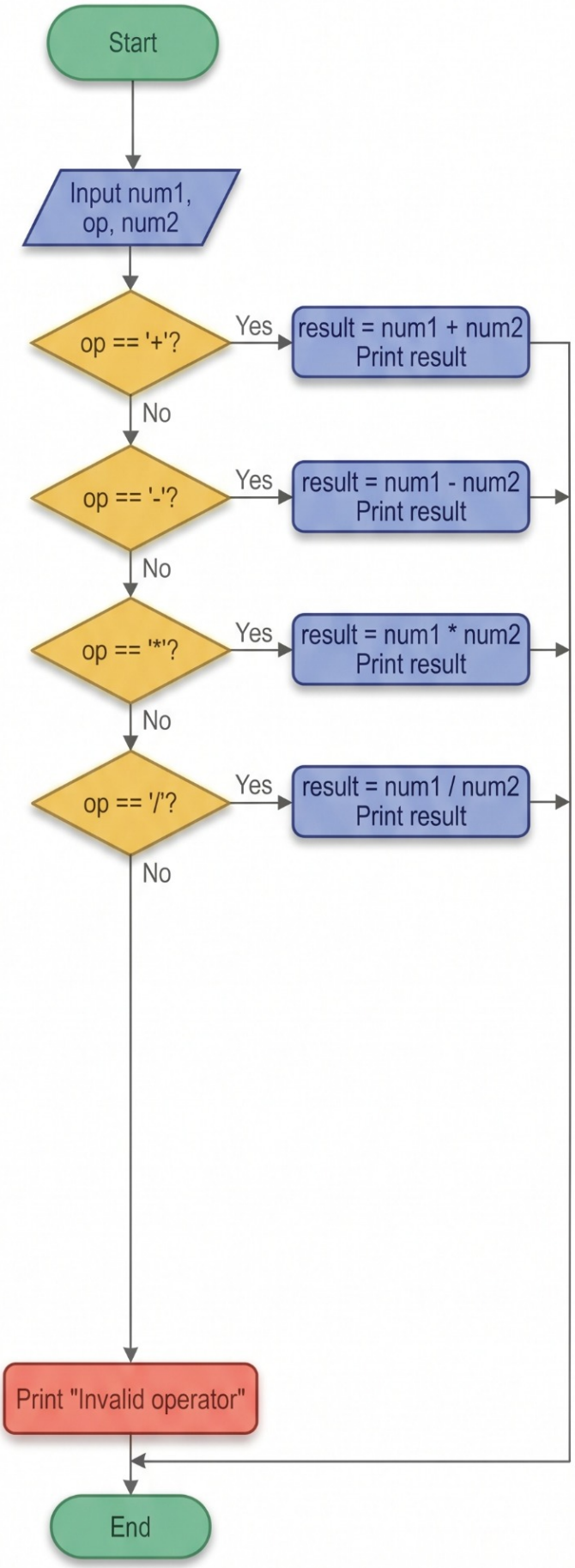
            //this (else if) is used for the subtraction operation whenever we want to subtract the numbers
            else if (sign == "-")
            {
                sub = a - b;
                Console.WriteLine("Subtraction is:" + sub);
            }

            //this (else if) is used for the multiplication operation whenever we want to multiply the numbers
            else if (sign == "*")
            {
                mul = a * b;
                Console.WriteLine("Multiplication is:" + mul);
            }

            //this (else if) is used for the division operation whenever we want to divide the numbers
            else if (sign == "/")
            {
                div = a / b;
                Console.WriteLine("Division is:" + div);
            }

            //this (else) is used when we input the invalid sign or number
            else
            {
                Console.WriteLine("Invalid sign or Number");
            }
        }
    }
}
```

# Flowchart


Start
Input num1, op, num2
op == '+'?
Yes
result = num1 + num2
Print result
No
op == '-'?
Yes
result = num1 - num2
Print result
No
op == '*'?
Yes
result = num1 * num2
Print result
No
op == '/'?
Yes
result = num1 / num2
Print result
No
Print "Invalid operator"
End

# Program Output: Calculator

